Lab Report

Summer 2025

Course Code: **EEE203L**

Course Title: **Electronic Circuits II Laboratory**

Lab Section:**02**

Experiment No.: 2

Name of the experiment: verification of KVL in AC circuit

Date of Submission:

Group No: **3**

Group Member Details:

1.	*Name:*	**Souvik Barman Ratul**
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	I.D.	**24121268**
3.	*Name:*	**Mahdi Zaman**
	I.D.	**24121307**
4.	*Name:*	**Alif Tamjid**
	I.D.	**24121308**

Submitted By :

Name : Souvik Barman Ratul

Submitted To :

Saad Mahbub Chowdhury

I.D. : 24121205

Compilation of result and Calculations (with ckt diagram and verification of result):

Circuit Diagram:

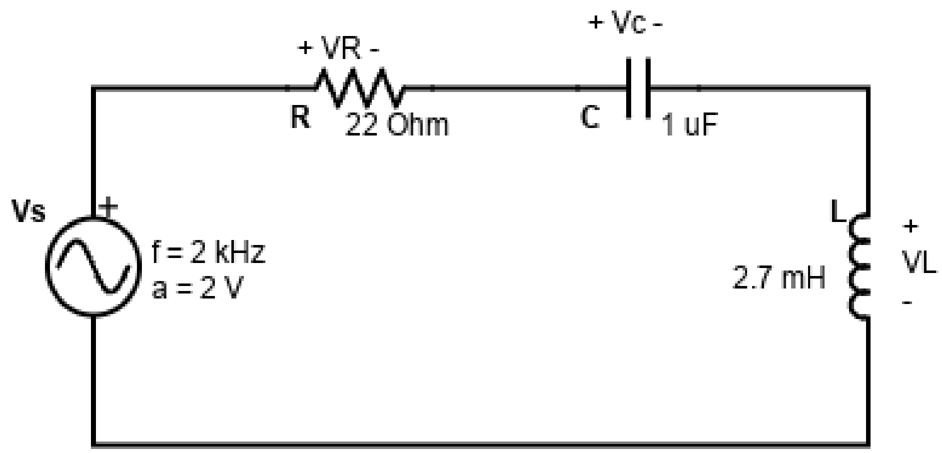


Figure 1

Data Sheet :

Table For V_S

V_{Sa} (V) (Peak value from Oscilloscope	$\lvert V_s \rvert = V_{Sa} / \sqrt{2}$ (V)	F (kHz) (From Oscilloscope)	$\angle V_s$ (0)
2	1.414	1.99	0

Table For V_R

V_{Ra} (V) (Peak value from Oscilloscope)	$\lvert V_R \rvert = V_{Ra} / \sqrt{2}$ (V)	Sign (+/-)	Δt (ms) (From Oscilloscope)	F (kHz) (From Oscilloscope)	$360f\Delta t$ (0)	$\angle V_R$ (0)
0.74	0.523	(+)	0.078	1.99	55.879	+55.879

Table For V_C

V_{Ca} (V) (Peak value from Oscilloscope)	$\lvert V_C \rvert = V_{Ca} / \sqrt{2}$ (V)	Sign (+/-)	Δt (ms) (From Oscilloscope)	F (kHz) (From Oscilloscope)	$360f\Delta t$ (0)	$\angle V_C$ (0)
2.62	1.853	(-)	0.206	1.99	147.578	-147.578

Table For V_L

V_{La} (V) (Peak value from Oscilloscope)	$\lvert V_L \rvert = V_{La} / \sqrt{2}$ (V)	Sign (+/-)	Δt (ms) (From Oscilloscope)	F (kHz) (From Oscilloscope)	$360f\Delta t$ (0)	$\angle V_L$ (0)
1.07	0.7566	(+)	0.180	1.99	128.95	+128.95

Table For $V_R + V_C + V_L$

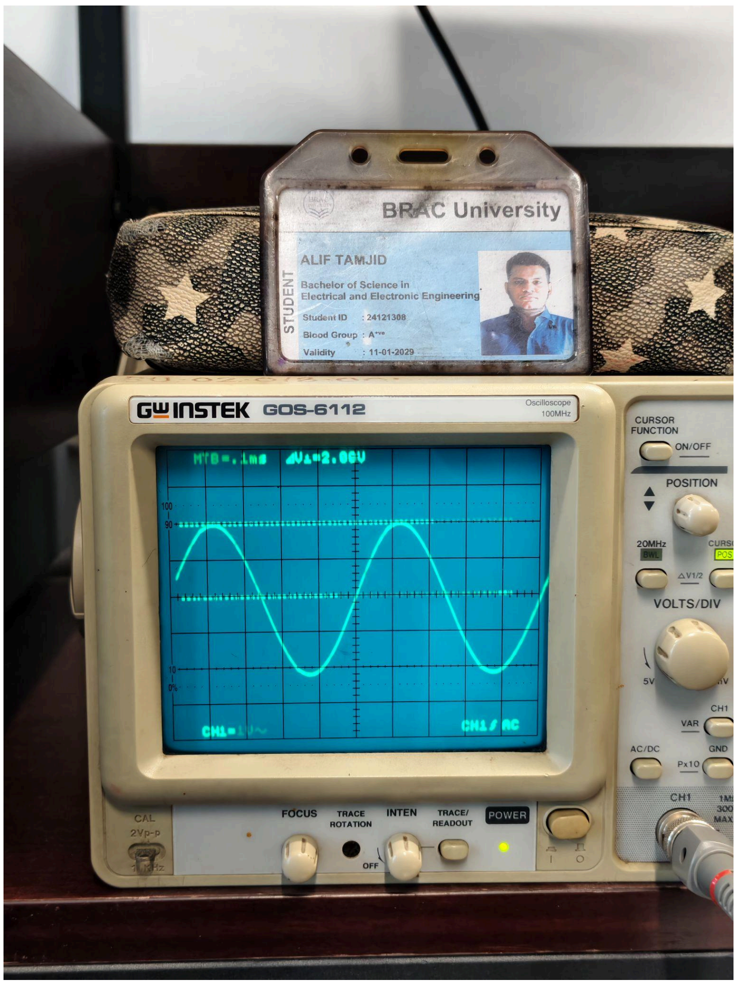
$\lvert V_R + V_C + V_L \rvert$ (V)	$\angle V_R + V_C + V_L$ (0)
(0.523+1.853+0.7566) =3.1326	(+55.879-147.578+128.95) =37.251

Error $= \{ (\lvert V_R + V_C + V_L\rvert - \lvert V_S \rvert) / (\lvert V_R + V_C + V_L \rvert) \}$ ✖ 100%

$= \{ (3.1326 - 1.414) / 3.1326 \}$ ✖ 100%

$= 54.86\%$

Classwork Proof :

<table>
<tr><th>V_S</th><th>V_R</th></tr>
<tr><td>


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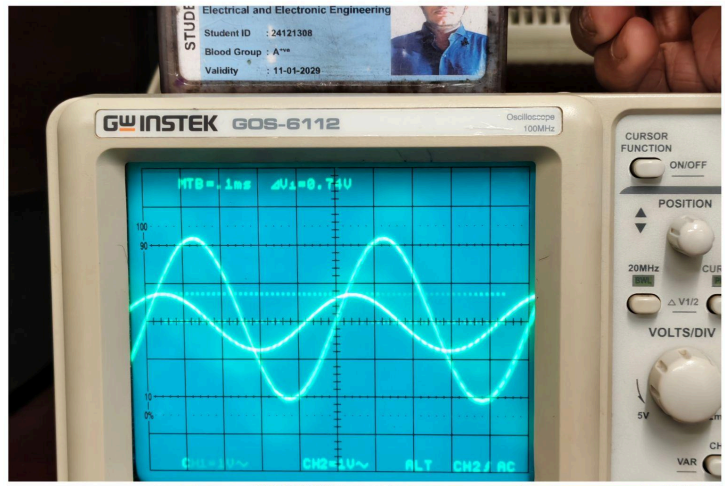
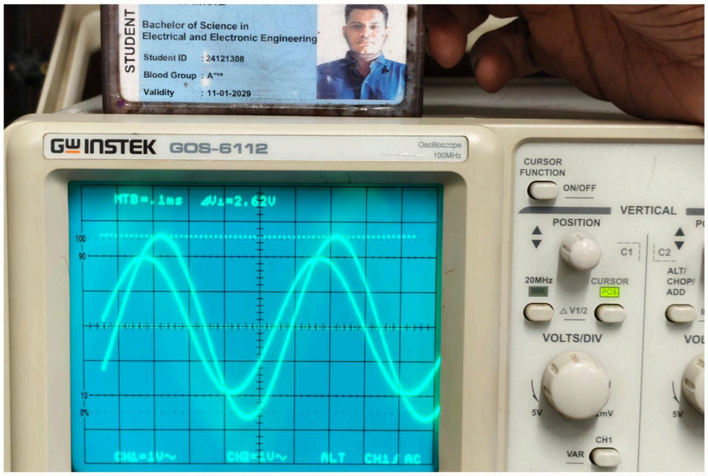
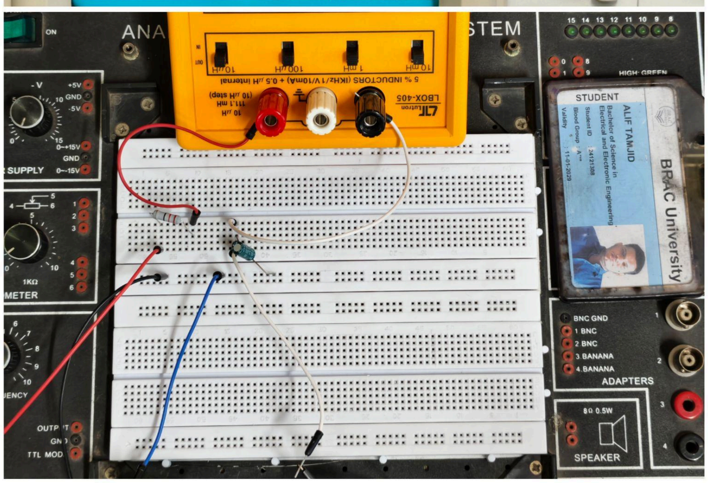


</td></tr>
</table>

<table>
<tr><th>V_C</th><th>V_L</th></tr>
<tr><td>




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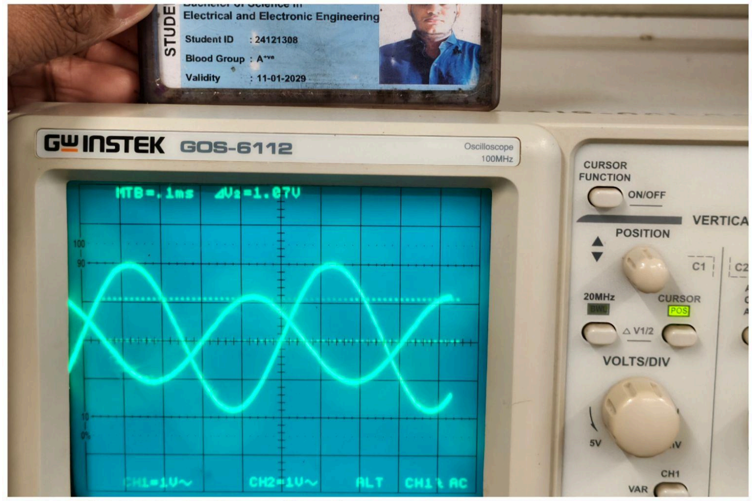

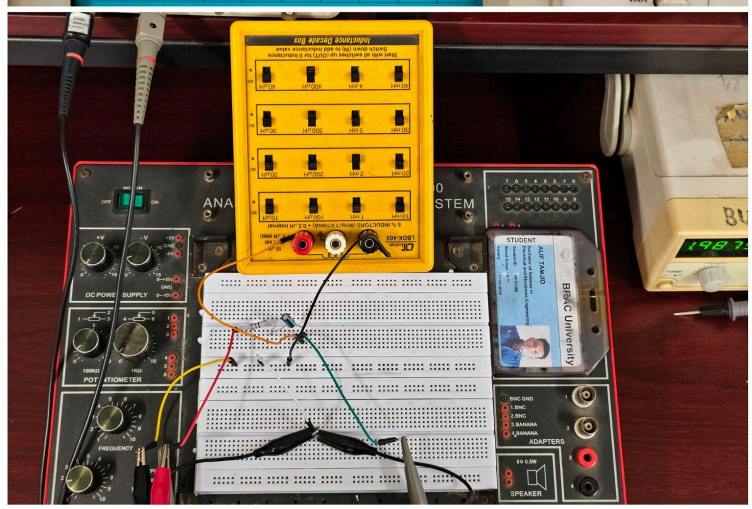

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Appendix:

Brac University
Department of Electrical & Electronic Engineering (EEE)
EEE203L – Electrical Circuits II Laboratory

Experiment 2

Name of the Experiment:

Verification of KVL in AC Circuits

Objective:

The objective of this experiment is to study RLC series circuit when energized by an ac source. KVL in phasor form will also be verified.

Circuit Diagram:

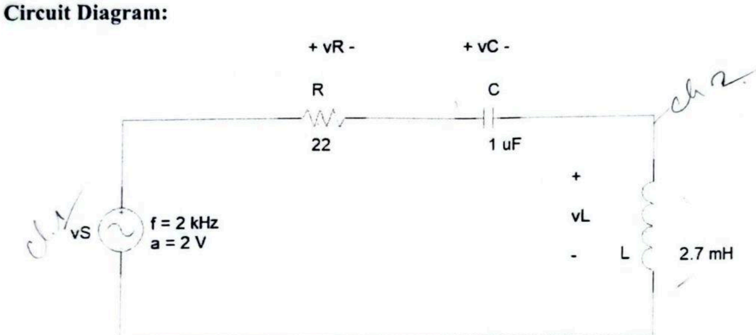

Figure 1 : KVL circuit

Equipments:

1. Multimeter
2. AC ammeter
3. Resistances – 22Ω (1piece) and 2.5kΩ(3 pieces)
4. Capacitor –1 uF (1 piece)
5. Inductor – 2.7 mH (1 piece)
6. Bread Broad
7. Wires for connection

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Procedure:

1. Construct the circuit as shown in figure 1.
2. Observe the wave shapes for the voltages of V_R, V_L, and V_C and fill up the required table.

Data:

Table for V_s			
V_{sa} (V) (Peak value from Oscilloscope)	$\lvert Vs \rvert = \frac{V_{sa}}{\sqrt{2}}$ (V)	f (kHz) (From oscilloscope)	$\angle V_s$ (o)
2 V	1.414	1.99	0

Table for V_R						
V_{Ra} (V) (Peak value from Oscilloscope)	$\lvert V_R \rvert = \frac{V_{Ra}}{\sqrt{2}}$ (V)	Sign (+/-)	Δt(ms) (From oscilloscope)	f(kHz) (From oscilloscope)	360fΔt (o)	$\angle V_R$ (o)
0.74	0.523	(+)	0.078	1.99	55.879	+52.879

Table for V_C						
V_{Ca} (V) (Peak value from Oscilloscope)	$\lvert V_C \rvert = \frac{V_{Ca}}{\sqrt{2}}$ (V)	Sign (+/-)	Δt(ms) (From oscilloscope)	f(kHz) (From oscilloscope)	360fΔt (o)	$\angle V_C$ (o)
2.62	1.853	(−)	0.206	1.99	147.578	−147.578

Table for V_L						
V_{La} (V) (Peak value from Oscilloscope)	$\lvert V_L \rvert = \frac{V_{La}}{\sqrt{2}}$ (V)	Sign (+/-)	Δt(ms) (From oscilloscope)	f(kHz) (From oscilloscope)	360fΔt (o)	$\angle V_L$ (o)
1.07	0.7566	(+)	0.180	1.99	128.95	+128.95

Table for $V_R + V_C + V_L$	
$\lvert V_R + V_C + V_L \rvert$ (V)	$\angle V_R + V_C + V_L$ (o)
3.1326	37.251

Saad
13/02/25

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Conclusion:

We built a circuit using a resistor, capacitor, and inductor all in a line. Then we sent wave-like electricity (AC) through it and measured how the waves changed in each part. We added up the voltages (and their wave angles) from all the parts and compared it to the total voltage from the source. We found that:

1. The total we got was close to what we expected.
2. There was some error (about 54.86%), maybe because of mistakes while measuring or because real parts do not behave perfectly.

So, even though we had some small differences, the experiment mostly matched what KVL said should happen. That means KVL works in AC circuits, just like in normal ones, but we need to be careful with phase angles.